

2 The Second Law of thermodynamics

The Second Law expresses in mathematical and physical terms the distinction between reversible and irreversible processes and in particular incorporates the empirical observation that heat flows from a hot body to a cold body and not the other way round (disregarding situations where a general relativistic description is needed!)¹ A very important consequence is an expression for maximum efficiency of a cyclic heat engine (or refrigerator). As a byproduct the concept of Entropy as a state function emerges which, when married with a microscopic description, endows the Second Law with additional physical insight.

Before stating the Second Law, we will note some useful notation and a mathematical result.

2.1 Preliminaries I: diagrams for heat engines

Remember that the adiabatic stages of a Carnot cycle have no heat transfer, $\Delta Q = 0$, and the work performed is equal in magnitude and opposite in sign, for the two adiabatic legs, so cancels. However the isothermal stages absorb net heat and perform net work – so are the crux of the cycle acting as a heat engine. In Fig. (2.1) on the left we represent the Carnot cycle in the p - V plane, and on the right there is a simplified depiction where the adiabatic stages are not mentioned and the focus is on the important ingredients. Such shorthand diagrams are very useful in establishing important results about the Second Law and facilitating discussion of more complex situations.

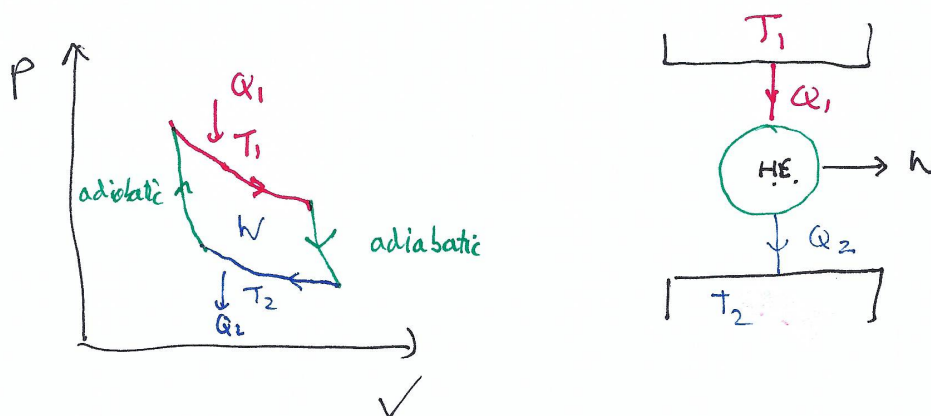


Figure 2.1: Left: the Carnot cycle indicating the isothermal (T_1 and T_2) and adiabatic stages. Right: a depiction of a Carnot cycle acting as a heat engine operating between two heat reservoirs at T_1 and T_2 .

¹See, for example, *Relativity: An introduction to Special and General Relativity* H. Stephani, Third Edition, CUP (2004), p 290-291.

2.2 Preliminaries II: how to turn Q into a function of state

Remember from Mathematics 1B if one wishes to ascertain whether, given the functions $M(x, y)$ and $N(x, y)$, the RHS of

$$\begin{aligned} df &\stackrel{?}{=} M(x, y)dx + N(x, y)dy \\ &\stackrel{?}{=} \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy \end{aligned} \quad (2.1)$$

really is a total differential of a smooth function, $f(x, y)$, then the mixed partial derivatives must be equal. I.e. if

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = \frac{\partial^2 f}{\partial x \partial y} \quad (2.2)$$

then Eq. (2.1) is indeed a total differential of a smooth function, $f(x, y)$. We will now apply this test to the consideration of Q as a function of state.

Consider the First Law for an *ideal gas* with specific heat, $C_v(T)$, and use the equation of state to replace p

$$\delta Q = dU + p dV = C_v dT + \frac{nRT}{V} dV. \quad (2.3)$$

If $Q = Q(T, V)$, were a function state (the dependence on T and V is indicated by the differentials on the RHS of Eq. (2.3)), then Eq. (2.3) would be a total differential as in Eq. (2.1) above

$$\left(\frac{\partial Q}{\partial T} \right)_V \stackrel{?}{=} C_v \quad \text{and} \quad \left(\frac{\partial Q}{\partial V} \right)_T \stackrel{?}{=} \frac{nRT}{V}.$$

However, using Eq. (2.1):

$$\frac{\partial^2 Q}{\partial V \partial T} = \frac{\partial C_v}{\partial V} = 0 \quad \text{and} \quad \frac{\partial^2 Q}{\partial T \partial V} = \frac{nR}{V} \neq 0 \quad \Rightarrow \quad \frac{\partial^2 Q}{\partial V \partial T} \neq \frac{\partial^2 Q}{\partial T \partial V},$$

and so Q is **not** a function of state – the history is important.

However, now divide both sides by T and denote, optimistically, $\delta Q/T = dS$, a total differential:

$$\begin{aligned} \frac{\delta Q}{T} = dS &= \frac{C_v}{T} dT + \frac{nR}{V} dV \\ \Rightarrow \quad \frac{\partial^2 S}{\partial V \partial T} &= \frac{\partial}{\partial V} \frac{C_v}{T} = 0 \\ \text{and} \quad \frac{\partial^2 S}{\partial T \partial V} &= \frac{\partial}{\partial T} \frac{nR}{V} = 0 \\ \text{So:} \quad \frac{\partial^2 S}{\partial V \partial T} &= \frac{\partial^2 S}{\partial T \partial V}, \end{aligned} \quad (2.4)$$

implying that, indeed,

$$\frac{\delta Q}{T} = dS$$

a total differential of a function we will denote $S(T, V)$. This will turn out to be entropy in general.

2.3 The Kelvin and Clausius statements of the Second Law

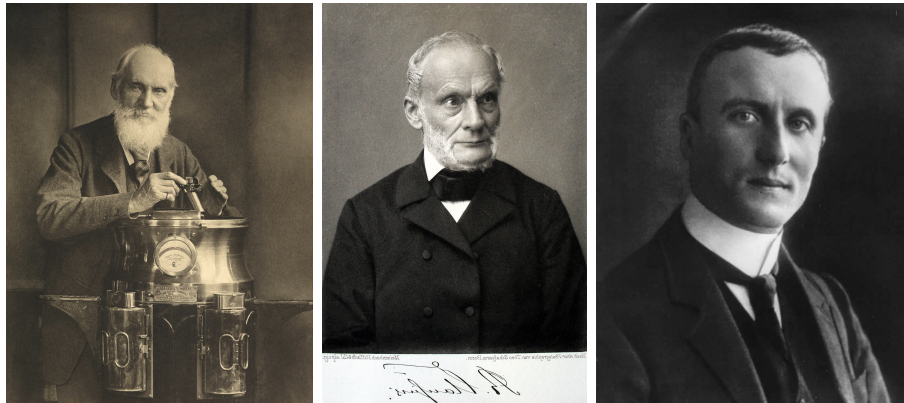


Figure 2.2: Left : Kelvin. Centre: Clausius. Right: Carathéodory. Credit: Wiki

There are two straightforward statements of the Second Law and a third² which is both simpler and more abstract. We focus on the first two, due to Kelvin and Clausius.

The Kelvin Statement

It is impossible to devise an engine which, working in a cycle, shall produce no effect other than the extraction of heat from a reservoir and the performance of an equal amount of work.

The Clausius statement:

It is impossible to devise an engine which, working in a cycle, shall produce no effect other than the transfer of heat from a colder to a hotter body.

The insistence on a cyclic process is important. For Clausius' formulation, one could consider expanding a gas in a cylinder in thermal contact with the colder body. Now expand the gas so that it extracts heat from the colder body. Then compress the gas until hotter than the hot body and let it donate heat to the hot body. But this is not a cycle as its final state – the temperature of the hot body – was not its initial state, the temperature of the colder body. thus it is not a cyclic process.



Figure 2.3: Disallowed heat engines: left: Kelvin, with no heat released at the low temperature reservoir; right: Clausius, with only heat transport from cold to hot and no work performed.

²The third, Carathéodory, see Adkins Ch 6.

We may see that each formulation of the Second Law implies the other. Consider Fig. (2.4) a composite engine made from a Kelvin-impossible engine (taking heat from hot reservoir and delivering work with no heat deposited at low temperature) feeds work into a Carnot cycle working in reverse as a refrigerator/heat pump. The net result, noting the heat removed from the hot reservoir cancels - as it is added back - so the net result is to transport heat from the cold reservoir and donate it to the hot reservoir with no work performed on external bodies – violating the Clausius statement.

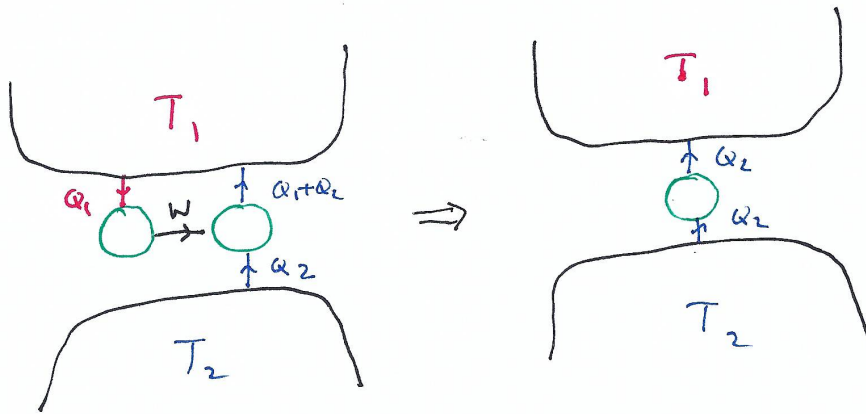


Figure 2.4: Left: An impossible Kelvin engine feeding a refrigerator or heat pump. Right: The composite is equivalent to an impossible Clausius engine.

One may establish the inverse result – see problem (2.2).

2.4 Efficiency of heat engines

One motivation for the early work on thermodynamics was to develop better steam engines. "Better" could mean minimising the amount of heat provided in generating work, as the upper thermal reservoir, at T_1 , needs to be created by burning fuel (the lower one, T_2 , is usually as close to room temperature as possible). So the financial cost per Joule of work extracted was an impetus to the subject.

A direct measure of "efficiency", η , is the ratio of work from system to the heat input at T_1 – the input from fuel burning.

$$\eta = \frac{W_{\text{by}}}{Q_{\text{in}}} = \frac{Q_1 - Q_2}{Q_1} \quad (\text{For Carnot cycle.})$$

$$= 1 - \frac{Q_2}{Q_1} \quad (2.5)$$

$$= 1 - \frac{T_2}{T_1} . \quad (2.6)$$

To approach unity (i.e. almost perfect conversion of heat into work, the limit disallowed by Kelvin formulation), either the hot reservoir approaches infinite temperature (expensive), or

the low temperature approaches absolute zero temperature, which would have to be achieved artificially, at a cost in refrigeration (see below) diminishing efficiency. The hot limit is related to the research done in the School of Metallurgy and Materials Science. They work on development of alloys to allow aircraft jet engines run as hot as possible for fuel economy (without melting the turbofan blades), although the engines are not ideal Carnot cycles. See [Adkins](#), p. 64-67, for a discussion of more realistic engines.

2.4.1 Refrigeration

Since Carnot cycles are reversible, they can be run "anticlockwise" to remove heat at T_2 and donate heat to T_1 . Here the low temperature "reservoir" could be food in a fridge and the high temperature "reservoir" be a kitchen. The heat is deposited via the cooling fins at the back of the fridge. In most domestic refrigeration the gas undergoing the cycle is $\text{CF}_3\text{CH}_2\text{F}$, 1,1,1,2-Tetrafluoroethane, or R-134a, substantially less risky in terms of the Ozone layer and climate warming than the older materials, CCl_2F_2 , "CFC's". Most refrigeration cycles involve a gas-liquid phase transition, so again a bit more involved than an ideal gas.

To have a "figure of merit" for refrigeration, η_R , a natural quantity is the ratio of the heat removed from the low-temperature reservoir to the work performed (by the fridge engine), i.e. the "cost"

$$\eta_R = \frac{Q_{\text{out}}}{W_{\text{on}}} = \frac{Q_2}{Q_1 - Q_2} = \frac{T_2}{T_1 - T_2} . \quad (2.7)$$

Note η_R may be greater than unity. Refrigeration becomes *very hard* as the low temperature reservoir $T_2 \rightarrow 0$; this is one reason why low-temperature physics is difficult!

2.4.2 Heat pumps

Were invented by Kelvin and are similar to fridges (reversed Carnot cycles) but the emphasis is on the delivery of heat to the hot reservoir, not the removal of heat from the cold reservoir. E.g. the heat delivered to the inside of a house from the outside. Thus the figure of merit is the heat delivered over the work performed by the electric motor driving it.

$$\eta_P = \frac{Q_1}{Q_1 - Q_2} = \frac{T_1}{T_1 - T_2} = 1 + \eta_R . \quad (2.8)$$

Note that the heat pump is most beneficial as the temperatures approach each other – when it is not needed! In the limit that $T_2 \rightarrow 0$, $\eta_P \rightarrow 1$, so there is no advantage to using the work done by its motor to directly heat the house. In between it can be useful: if $T_1 = 320$ K and $T_2 = 280$ K, 8 kW of heat comes from a power consumption of 1 kW, a reasonable advantage. A *ground-source* heat pump, uses heat from (ideally) more than 20 m underground. This depth ensures the ground temperature does not change significantly with seasons. Then the desired temperature inside the building and that in the ground are nearer to equality and hence the

figure of merit good on cold days. A problem with such heat pumps is the density of houses, unless either large suburban gardens or blocks of flats.

2.5 A Carnot cycle is most efficient

We will now see that the Carnot engine is the most efficient engine to run between two thermal reservoirs.

Consider a Carnot cycle, using an ideal gas, operating between T_1 and T_2 , with $T_1 > T_2$, with Q_1^c taken from the high temperature reservoir and Q_2^c delivered to the low temperature reservoir, so the work done is $W^c = Q_1^c - Q_2^c$. A second assumed **more efficient** cycle, D , also operates between the reservoirs accepting Q_1^D and donating Q_2^D , with associated work $W^D = Q_1^D - Q_2^D$. We may adjust the Carnot cycle so that $W^c = W^D = W$, by adjusting the number of moles in, or initial or final volumes of, the gas undergoing the Carnot cycle (remembering that $W \propto n \ln(V_{\text{initial}}/V_{\text{final}})$).

The efficiencies are

$$\eta^D > \eta^c \Rightarrow \frac{W}{Q_1^D} > \frac{W}{Q_1^c} \Rightarrow Q_1^c > Q_1^D. \quad (2.9)$$

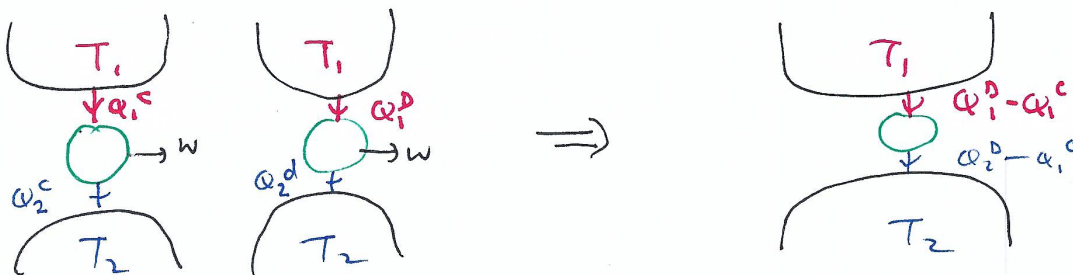


Figure 2.5: Left: The separate Carnot cycles C and D , with identical work. Right: After combination with the Carnot cycle reversed.

Then let us operate the Carnot cycle in reverse (we may as it is reversible), absorbing the work, W , from process D , to act as a refrigerator. Now, combine them into one heat engine, see right hand figure in Fig. (2.5). For this combined engine we see that energy is **donated to** the hot reservoir, $Q^{\text{tot}} = Q_1^D - Q_1^c$, as Eq. (2.9) states for the cycle D to be more efficient, $Q_1^c > Q_1^D$. By energy conservation (First Law) the same is true with regard to the energy removed from the cold reservoir. **No work** is performed.

Hence the assumption there exists a more efficient engine than an ideal gas Carnot engine

contradicts the Clausius statement of the Second Law and hence is impossible. Amongst the implications is that changing the gas from being ideal will not be more efficient.

If we consider the second (two-temperature) cycle to be *reversible*, with efficiency η_{rev} , we could, by driving it backwards, deduce that $\eta_{\text{rev}} \geq \eta_{\text{C}}$. For both to be true all reversible engines must have the same efficiency as the Carnot cycle, i.e. $\eta_{\text{rev}} = \eta_{\text{C}}$.

Could having an intermediate, third, temperature reservoir lead to a more efficient heat engine? See Problem (2.3).